Physical motor-to-sensor channels | z = 0 mm

Independent phase unwrap; one constant branch shift per curve. Same-record development check.

Same-record held-out development FRF — Additive SRIVC baseline — Fixed-pole structure + channel-delay refinement

$u_1 \to y_1$ $u_2 \to y_1$ $u_3 \to y_1$ $u_4 \to y_1$

$u_1 \to y_2$ $u_2 \to y_2$ $u_3 \to y_2$ $u_4 \to y_2$

$u_1 \to y_3$ $u_2 \to y_3$ $u_3 \to y_3$ $u_4 \to y_3$

$u_1 \to y_4$ $u_2 \to y_4$ $u_3 \to y_4$ $u_4 \to y_4$

Unwrapped phase (deg)

Frequency (Hz)